

Utilizing the Inception-ResNetV2 Pre-trained Model for Binary Classification of Leukemia Cells: An Advanced Approach to Hematological Diagnostics

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Abstract— Leukaemia, a form of hematologic malignancy, affects individuals across all age groups and is among the primary contributors to global mortality. Acute lymphoblastic leukaemia (ALL) is a prevalent form of leukaemia that mostly affects the bone-marrow in humans. The conventional methods employed for disease diagnoses, such as blood and bone marrow exams, are characterized by their sluggish and painful nature. To stop leukaemia from killing people, creating a computer-based automatic and strong classification system has become important. Algorithms developed using deep learning are widely utilized in contemporary medical practice for leukaemia treatment, namely in the identification and diagnosis of leukaemia in patients. This research presents an advanced approach in deep learning for the detection and categorization of leukaemia cells. To carry out this analysis, a Convolutional Neural Network model utilizing the Inception_Resnet50 architecture has been suggested. The Leukaemia Classification dataset, which contains 15114 images, is utilized to train and evaluate the model. The model has shown a very good accuracy of 95.75% in classifying the cells for ALL and Normal classes. This research can be implemented in the healthcare field for pre-diagnosis of Leukemia in blood cells.

Keywords— Leukemia detection, Inception Resnet50, Feature Extraction, Transfer Learning, Blood Cell examination

I. INTRODUCTION

Acute lymphocytic leukaemia (ALL) is a blood-related illness that spreads quickly throughout the body and it can result in catastrophic effects if untreated and undetected. One of the key phases entails inspecting samples of peripheral blood to examine white blood cells to ensure the early recognition of ALL and, subsequently, the chances of healing [1]. Leukaemia is a group of cancers that start in the bone marrow and cause the blood to have too many immature leukocytes. The word "leukaemia" comes from the Greek words "leukos," which means "white," and "haima," which indicates "blood" [2]. Leukaemia is not just one disease, but a group of linked diseases that affect both children and adults. It is a big problem for public health. Leukaemia is caused by a mix of genetics and the surroundings. Gene changes can cause cells to grow out of control since they don't have to go through the normal checks and balances that keep cellular division under control [3]. Some known risk factors include exposure to radiation or certain chemicals, a family history of leukaemia, or a genetic disease that affects your blood. Even though medicine has

come a long way, it is still hard to find leukaemia in its early stages. Examinations of the blood, bone marrow biopsy procedures, and cytogenetic analysis are the traditional ways to find out what's wrong [5]. Traditional diagnostic techniques for blood cancer involve manual approaches, wherein a pathologist examines a human blood sample using a microscope. One prominent concern that arises is the problem of inaccurate diagnostic procedures [6]. It is a well-established fact that humans, even medical professionals, are prone to errors. Consequently, it is not uncommon for doctors to make mistakes, leading to inaccurate diagnoses that tragically contribute to the loss of thousands of lives every.

In the past few years, combining machine learning and deep learning has shown a lot of potential for making leukaemia diagnoses more accurate and faster [7]. With the development of deep learning and other advanced computer methods, it is possible to make diagnostic tools that are faster, more accurate, and less invasive. By combining these new methods, it is possible to label and treat leukaemia in a much better way, which would be good for patients in the long run.

In this study, an Inception ResNetV2-based CNN model has been employed for the classification of blood cells for normal and leukaemia classes. This study work investigates the utilization of the Inception-ResNet50 pre-trained model for the purpose of binary classification of leukaemia cells. Through the utilization of this sophisticated model, our objective is to make a valuable contribution to the enhancement of diagnostic methodologies in the field of leukaemia, which has the potential to result in more accurate and prompt interventions. For training this model Leukemia classification dataset is utilized, a medical dataset available publically. With the implementation of this proposed model, the detection of leukemia cells will become more accurate.

II. LITERATURE REVIEW

Leukaemia is a potentially fatal medical condition that impacts the functionality of white blood cells. To achieve autonomous detection, a convolutional neural network (CNN) was utilized. Specifically, a pre-trained AlexNet model was implemented, wherein the previously trained neural network layers were substituted with newly refined layers. The categorization of acute lymphoblastic leukaemia and its various subtypes into four distinct classes is done by

[8]. The model utilizes robust partitioning and advanced learning techniques, specifically employing a convolutional neural network, to train bone marrow images to get accurate characterization outcomes. The authors of [9] established a framework that utilizes mobile-cloud assistance to effectively split and classify leukocytes into five distinct types. The initial step of the architecture was the segmentation of WBC using a color k-means technique. This technique effectively eliminated extraneous components by the use of morphological procedures. Ultimately, the process of classification was executed using an ensemble multi-class support vector machine (SVM). The automatic detection mechanism for acute leukaemia was presented by the authors of [10]. The initial step of the method involved the initial processing of distortion and smearing in digital images captured using microscopic means. Two classification models, namely k-nearest neighbor (K-NN) and naïve Bayes, were employed. The results obtained from conducting experiments on a dataset consisting of 60 blood samples demonstrate the higher performance of the K-NN classifier, as evidenced by its classification accuracy of 92.8%. The categorization scheme presented by the authors in reference [11] distinguishes between two distinct forms of acute leukaemia, namely acute myelogenous leukaemia (AML) and ALL. A total of twelve distinct features were retrieved from the image samples through manual means. In the present study, a K-NN classifier was utilized for classification. The experimental analysis conducted on a dataset including 1500 images yielded an accuracy rate of 86%. Authors of [12] established an AML classification approach that improved image contrast and identified five distinct characteristics. The categorization task was executed using an SVM classifier. The classification success rate of 93.5% was achieved through experiments conducted on a dataset consisting of 50 images. The authors in reference [13] introduced an AML system for classification that utilizes k-means and contour signatures methodologies to separate nuclei. Subsequently, the process of feature extraction, namely for cell size and cell color, was conducted using morphological techniques. The results of the conducted trials on a dataset including 100 photos indicate that the classifier using the SVM exhibited an accuracy rate of up to 92%. The automated technique for diagnosing ALL was provided by the authors in reference [14]. The white blood cells (WBCs) were segmented using histogram equalization and the Zack method. Subsequently, a variety of characteristics such as color, shape, texture, and mixed properties were retrieved. Next, the social spider optimization method was utilized to identify the most prominent traits. In conclusion, a variety of classifiers were employed for classification. The methodology was assessed using a dataset including 260 photos. The K-NN classifier yielded the most favorable outcomes with an accuracy rate of 95.67%. In their study, authors of [15] introduced a categorization approach for white blood cells (WBCs) utilizing multiple deep-learning networks. The system underwent evaluation using a dataset consisting of 2000 microscopic images depicting seven distinct types of white blood cells (WBCs). A comparative analysis was conducted against various conventional approaches. The findings collected demonstrated the higher performance of employing a Convolutional Neural Network (CNN), as evidenced by an accuracy rate of 88.5%.

The proposed technique attained an estimated categorization accuracy of 96.6% when assessed using a dataset including 1188 photos. The classification of ALL and its subtypes was conducted by the authors of [16] by the utilization of a previously trained AlexNet. For the evaluation, a data set known as ALL-IDB2, consisting of a total of 260 pictures, was utilized. The obtained data demonstrated an average accuracy for classification of 96.06%. This study proposes an advanced CNN model based on Inception ResnetV2. The Leukaemia classification dataset has been utilized for conducting this classification task. The model has shown very good results in this binary classification for leukemia cells.

III. MATERIALS AND METHODS

This study utilizes the transfer learning approach for classifying the WBCs for normal and leukemia classes. This section will describe the dataset, image processing, proposed methodology and experimental setup which is followed to conduct this study.

A. Dataset Description

The Leukemia Classification dataset has been utilized for the training and testing of the proposed model. This dataset includes the 15,114 segmented microscopic images of 118 patients. This includes the acute lymphoblastic images (ALL) and HEM for normal cell images. The data set is divided into testing, training and validation sets. Fig.1 gives the details of the distribution of the images in which the total number of images in all three sets is given.

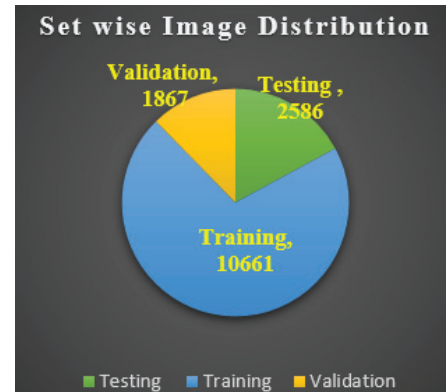


Fig. 1. Set Wise Image Distribution in the Dataset

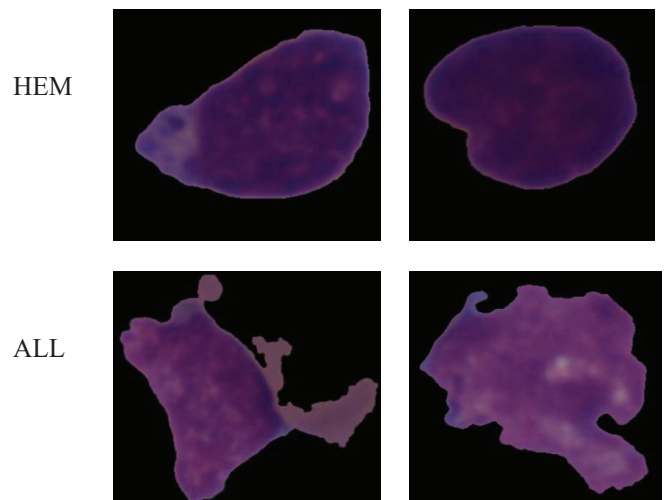


Fig. 2. Sample Images for Both Classes

Fig.2 gives the sample images of Leukemia (ALL) and HEM for normal and healthy cell classes from the dataset.

B. Image Pre-Processing

The visual appeal of microscopic images is contingent upon several factors, including the type of camera and microscopes used, the amount of light source employed, the angle at which the images are captured, variations in illumination, as well as the presence of noise [17]. So before feeding the images to the model, it becomes extremely important to make the images noise-free. Images should be of the same size, no color contradiction should be there. Therefore image preprocessing step is a very crucial step that has its value in the working of the model. Fig.3 gives all the steps which are followed for the preprocessing of the images. For this study, image preprocessing includes image resizing, color mode, shuffling of images, data augmentation, and scalar preprocessing function [18].

- 1) *Image Resizing*: In this method, images of different sizes are guaranteed to be standardized to a uniform shape of 224x224, before being fed into the neural network.
- 2) *Color Mode*: The color parameter indicates that the images are handled in the 'RGB' color mode. This indicates that the images are handled as three-channel, colored images (Red, Green, Blue).
- 3) *Shuffling of Images*: To contribute to a more effective training of the model, this phase involves randomly shuffling the labels and images before each epoch.
- 4) *Data Augmentation*: This stage solely involves the augmentation of the training data; the only alteration that is provided is the flipping of the horizon horizontally. This effectively doubles the variance in the training data by allowing photos to be randomly flipped horizontally during training. By injecting such unpredictability, it improves the model's ability to generalize.
- 5) *Scalar Preprocessing Function*: Scalar is a customized preliminary processing function that is defined and returns the input image. It doesn't carry out any extra normalization or transformation in this particular instance. With this feature in place, though, it may simply add any further preprocessing steps later on if needed.

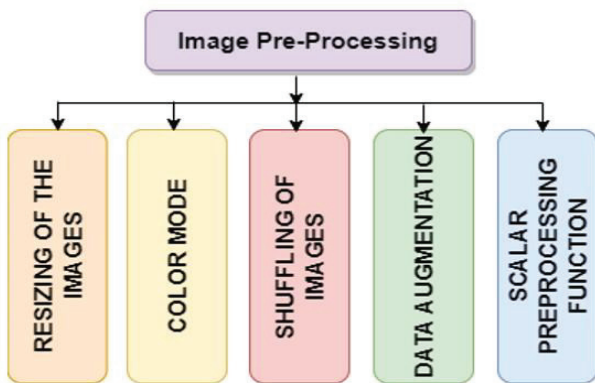


Fig. 3. Image Pre-Processing Steps

C. Proposed Methodology

For the binary classification of WBC, a CNN model based on Inception_Resnet_V2 architecture. In this study, a classification model is introduced to differentiate between microscopic images that portray healthy tissue and those that indicate leukaemia. This model utilizes TL by using Inception_Resnet_V2 pre-trained deep neural networks [19]. TL is a technique that reduces the time and effort required for designing and training networks from the beginning. The leukemia classification dataset has been employed for training this model. This dataset contains a total of 15,114 images for the normal and leukemia classes. Fig.4 gives the step representation of the proposed framework.

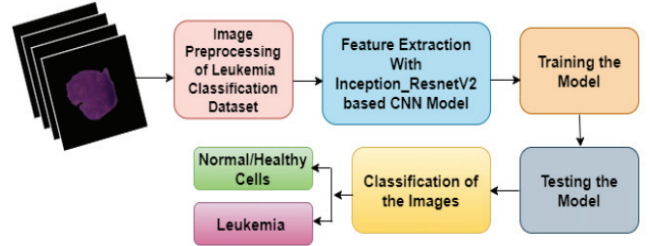


Fig. 4. Block Representation of Proposed Methodology

From fig.4 it can be seen that after the input images are fed to the network, preprocessing steps take place which include different steps which are shown in fig.3 After this the feature extraction process takes place which is conducted with the implication of the Inception_Resnet_V2-based CNN model. For training and testing of the proposed framework, a medical dataset named Leukemia classification [20] has been employed. This dataset is a collection of microscopic images of the patient's cells.

Because deeper networks are generally acknowledged to produce greater comprehension of the image, the network's structure has 164 layers. Therefore for this classification task, Inception_Resnet_V2 has been employed. Fig.5 gives the Inception_Resnet_V2 pre-trained model architecture.

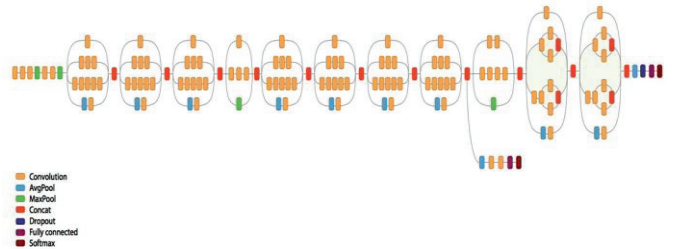


Fig. 5. Google's Inception_Resnet_V2 Architecture [20]

The Inception-ResNet-v2 framework is a deep learning model with a hybrid approach that combines the concepts of the Inception networks with the ResNet. The objective of this fusion is to leverage the advantages of both architectures: the effective decrease of grid size and the extraction of features at several levels offered by Inception networks, along with the inclusion of identity shortcut connections from ResNet, which enable the training of deeper models without encountering issues related to vanishing gradients [21]. Fig.6 gives the proposed model summary in which the trainable and non-trainable parameters are given

Model: "sequential"		
Layer (type)	Output Shape	Param #
inception_resnet_v2 (Functional)	(None, 1536)	54336736
batch_normalization_203 (Batch Normalization)	(None, 1536)	6144
dense (Dense)	(None, 256)	393472
dropout (Dropout)	(None, 256)	0
dense_1 (Dense)	(None, 2)	514
Total params: 54,736,866		
Trainable params: 54,673,250		
Non-trainable params: 63,616		

Fig. 6. Parameters Structure of Proposed Framework

D. Experimental Setup

To identify normal and abnormal WBCs in images, a CNN TL model [3] is proposed in this work. A medical dataset for Leukaemia classification has been employed to train the model, and it contains 15114 photos split into two categories. Adamax optimizer [17] is used to fine-tune the model. It is used to adjust model parameters to minimize the model error. This research was carried out using the Google Colab platform [19], the different TensorFlow libraries and the Image Data generator for image pre-processing. The amount by which the model should be adjusted whenever the weights of the model are updated in light of the expected error has been set to 0.0001. The model was trained with a 40-batch size for 17 epochs. A dropout of 0.45 has been applied to prevent overfitting during the training of the model. The CNN model's training hyperparameters are displayed in Table. I.

TABLE I. HYPERPARAMETERS UTILIZED FOR OPTIMIZATION OF THE MODEL

S.No	Simulation Parameter Name	Value
1.	Learning Rate	0.001
2.	Batch Size	40
3.	Epoch	17
4.	Optimizer	Adamax
5.	Loss Function	Binary Cross Entropy
6.	Dropout	0.45

IV. EVALUATION OF THE FINDINGS

This section presents the results achieved through the utilization of the proposed model. The model is evaluated on the Leukemia classification dataset. When evaluating the proposed model, multiple performance metrics are considered, including precision, sensitivity, F1 Score, and accuracy. The model is trained for 17 epochs and the values of Accuracy and loss are recorded. Table. II gives the details of the accuracy and loss achieved during the training of the model.

TABLE II. TRAINING AND LOSS OUTCOMES FOR TRAINING AND VALIDATION

Epochs	Training Loss	the	Validation Accuracy	Validation Loss
1	5.42	84.99	5.22	82.55
4	1.55	92.61	1.47	85.30
7	0.56	95.21	0.74	89.93

10	0.25	97.62	0.42	93.55
13	0.16	99.04	0.27	95.31
15	0.13	99.20	0.36	93.87
17	0.12	99.20	0.32	95.75

From Table. II, it can be seen that the training loss exhibits a steady pattern of decrease across the epochs, declining from 5.42 during epoch 1 to 0.12 by epoch 17. This observation suggests that the model is through a process of learning and enhancing its ability to accurately represent the training data. In contrast, the training accuracy has seen a steady enhancement, commencing at 84.99% at epoch 1 and reaching 99.20% by epoch 17. By the thirteenth epoch, the model demonstrates a high level of proficiency in predicting events on the training set, as seen by its accuracy nearing 99%. The validation loss exhibits a downward trend, starting at 5.22 during epoch 1 and progressively decreasing to 0.27 by epoch 13. Nevertheless, there is a marginal increase observed at epoch 15, followed by a subsequent decrease by epoch 17. This implies that although the model is continuing to enhance its ability to make accurate predictions on new data, there may be slight variations in performance. The validation accuracy consistently improves with time, starting at 82.55% in epoch 1 and reaching a peak of 95.75% in epoch 17. A minor decline in performance is observed at epoch 15, with an accuracy of 93.87%. However, this decline is subsequently rectified, as evidenced by the recovery in accuracy by epoch 17. Fig.7 and 8 represent the graphical presentation of Loss and Accuracy outcomes for training as well as validation.



Fig. 7. Graphical Representation of Training and Validation Loss

From fig.7 it can be seen that both the losses are consequently decreasing and at the 14th epoch, both the validation and training losses are at the lowest level. Fig. 8 displays the training and validation accuracy for the proposed model.

From fig. 8 it can be seen that the training accuracy is frequently increasing whereas the validation accuracy shows fluctuations in the starting epoch but still at the last epoch the validation accuracy value reached 95.75 which signifies that the proposed model is capable of classifying the microscopic WBC images very precisely for normal and leukemia classes.

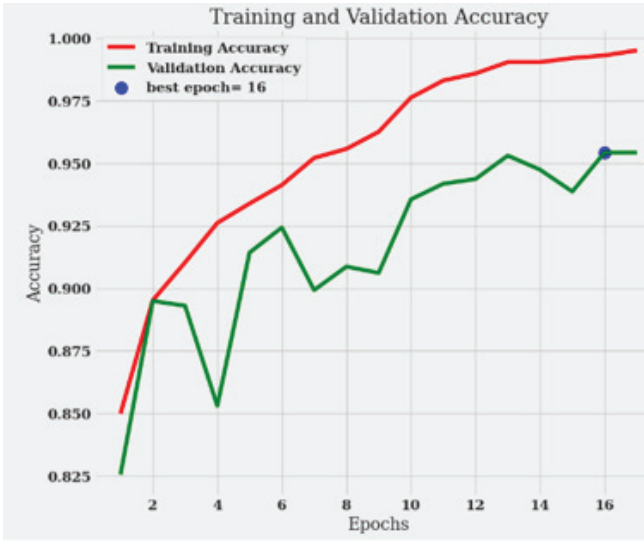


Fig. 8. Graphical Representation of Training and Validation Accuracy

A. Confusion Matrix

In this article, a CNN model based on TL has been presented for binary categorization of WBC to detect cases of Leukemia. The evaluation of the performance of these models is of utmost importance, and the confusion matrix emerges as an essential instrument in this assessment. To evaluate the efficacy of the methodology, the models's precision, recall, and F-1 score values have been recorded for both classes. The predictions made by the proposed model are shown in the confusion matrix, which compares them to the actual ground truth. Table.III gives the details of all the performance parameters for both classes.

TABLE III. PERFORMANCE ANALYSIS OF THE PROPOSED MODEL

S.No	Classes	Precision (%)	Recall (%)	f-1 score	Overall Accuracy
1	Normal	97.05	95.15	0.96	95.75
2	Leukemia	90.25	93.91	0.91	

Table.III presents a thorough analysis of the performance of a binary classification model that is entrusted with differentiating between "Normal" and "Leukemia" white blood cells. The model achieved a precision of 97.05% for the 'Normal' white blood cells, meaning that 97.05% of the predictions made for this class were correct. Furthermore, as indicated by its recall, it was able to accurately recognize 95.15% of the real "Normal" cases. For the 'Normal' class, the balance between these two criteria yields an F1 score of 0.96, which is noticeably high.

Conversely, the model's precision for "Leukemia" white blood cells is 90.25%, meaning that 90.25% of the cases that were predicted to be "Leukemia" were classified properly. With a recall of 93.91% for this class, it was able to accurately identify 93.91% of the real cases of "Leukemia" in the dataset. An F1 score of 0.91 for this area also reflects the harmony between precision and recall, which is noteworthy.

The model's overall classification accuracy, accounting for predictions for both groups, is 95.75%. This figure shows a strong capacity to accurately identify leukemia from normal white blood cells with a high degree of precision.

Fig.9 gives the confusion matrix for the proposed model performance description.

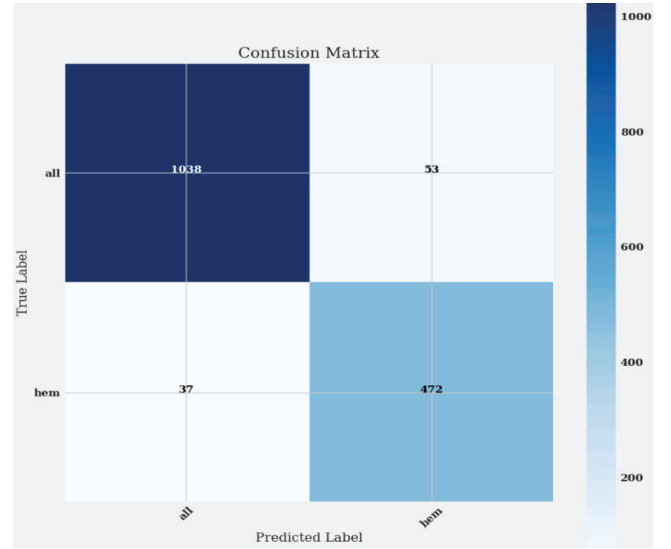


Fig. 9. Confusion Matrix for Proposed Model

B. Visualization of Results

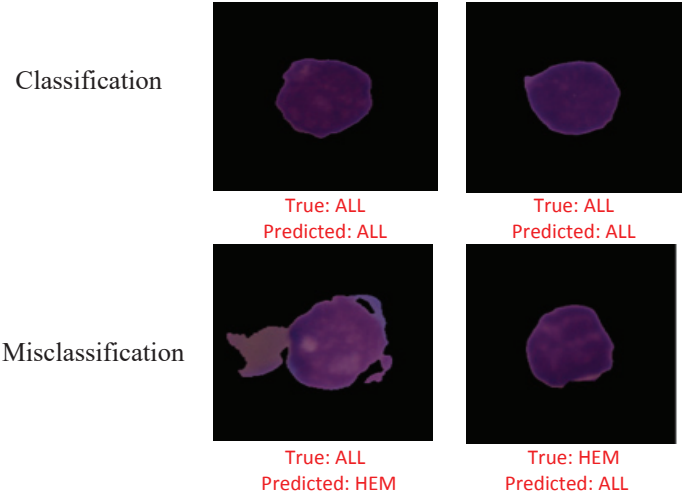


Fig. 10. Visualization of Results

V. CONCLUSION

The Inception_ResnetV2 architecture, a (CNN) model renowned for its deep complexity and adeptness at managing complicated picture patterns, served as the foundation for this work. The proposed model is trained over 17 epochs, and made use of the Leukemia classification dataset for training and testing of the model. This dataset includes a total of 15,114 pictures divided into the HEM (normal) and ALL (leukemia) classes. Preparing the model for optimal performance involved setting a batch size of 40 images, a learning rate of 0.001, and the Adamax optimizer to control the learning process. At 95.75% overall, the model demonstrated its ability to accurately classify the wide range of cell pictures. Turning to metrics unique to each class, the accuracy was 97.05% for the HEM class and 90.25% for the ALL class. The model's ability to decrease false negatives is further demonstrated by recall rates of 95.15% for ALL and 93.91% for HEM. An even performance in terms of recall and precision is shown by the computed F1 scores, which are 0.91 for ALL and 0.96 for HEM. The model can be further

utilized in the medical field for the categorization of Leukemia.8

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